

Suburban Insight

Helping Newcomers Choose Where to Live in Melbourne

DS01 · Project 7 · FIT3163 Data Science Project · Semester 1, 2026

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Overview

Hanjia Liu

#	Presenter	Topic	Slides
1	Hanjia Liu	Background, users, use cases and survey Gantt chart, conclusion	1–5, 15–16
2	Liuxiaoxi Yang	Requirements, solution, wireframe walkthrough	6–9
3	Jiajing Xu	Technical stack, feasibility, data, assumptions, testing	10–12
4	Aidan Kwok	PM methodology, WBS	13–14

Problem Background

Hanjia Liu

7.5M

people born overseas
in Australia

27.6%

of Australia's entire
population

71%

arranged
accommodation
before arriving in
Australia

52%

experienced at least
one housing-related
problem

THE CHALLENGE

- ***Where should I live?***
- Choosing suburb is difficult for newcomers
 - affordability
 - safety considerations and many more settlement related factors
- Current housing information is fragmented



Goal

To improve settlement decision-making for newcomers through a centralised suburb comparison platform

End Users & Use Cases

Hanjia Liu

PROJECT TARGET END USERS

- Primary User
 - International students newly arrived in Melbourne
- Secondary User
 - Newcomers with working visa or new migrants



USE CASES

- Use Case 1: A Chinese international student wants to find a suburb with: **affordable rent** and **Chinese-speaking residents near the university**
- Use Case 2: A migrant family wants to find suburbs by: **income levels, nearby services, and overall living conditions**



User Research / Survey Results

Hanjia Liu

20

international
students surveyed

90%

lack of suburb
knowledge & too much scattered
information

80%

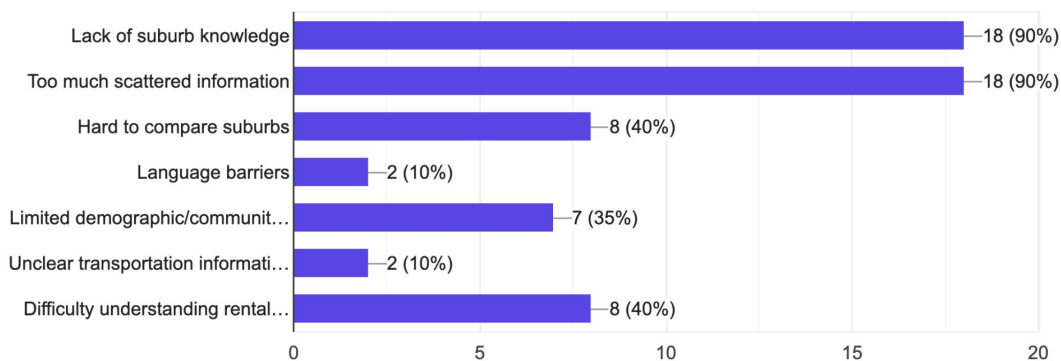
interactive suburb map

100%

would use our platform if
installed

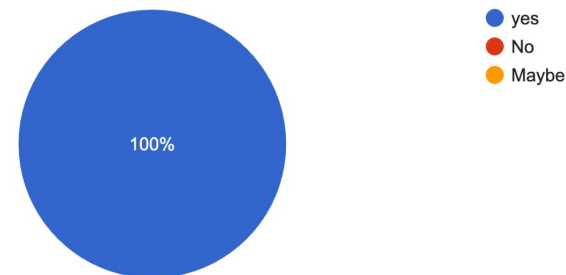
What difficulties did you experience when searching for accommodation in Australia?

20 responses



Would you use this type of platform in the future?

20 responses



5 Significant Requirements

Liuxiaoxi Yang

Req.1

FR, High

Interactive map with clickable suburbs showing demographics

Newcomers need a visual, intuitive way to explore unfamiliar areas

Req.2

FR, High

Demographic filters by rent, income, cultural background

Users need personalised narrowing that existing platforms do not offer

Req.3

FR, Med

ML suburb clustering using K-means

Newcomers may not know which suburbs share similar demographic profiles

Req.4

FR, Med

Side-by-side suburb comparison with charts

Users evaluating multiple suburbs need direct visual comparison

Req.5

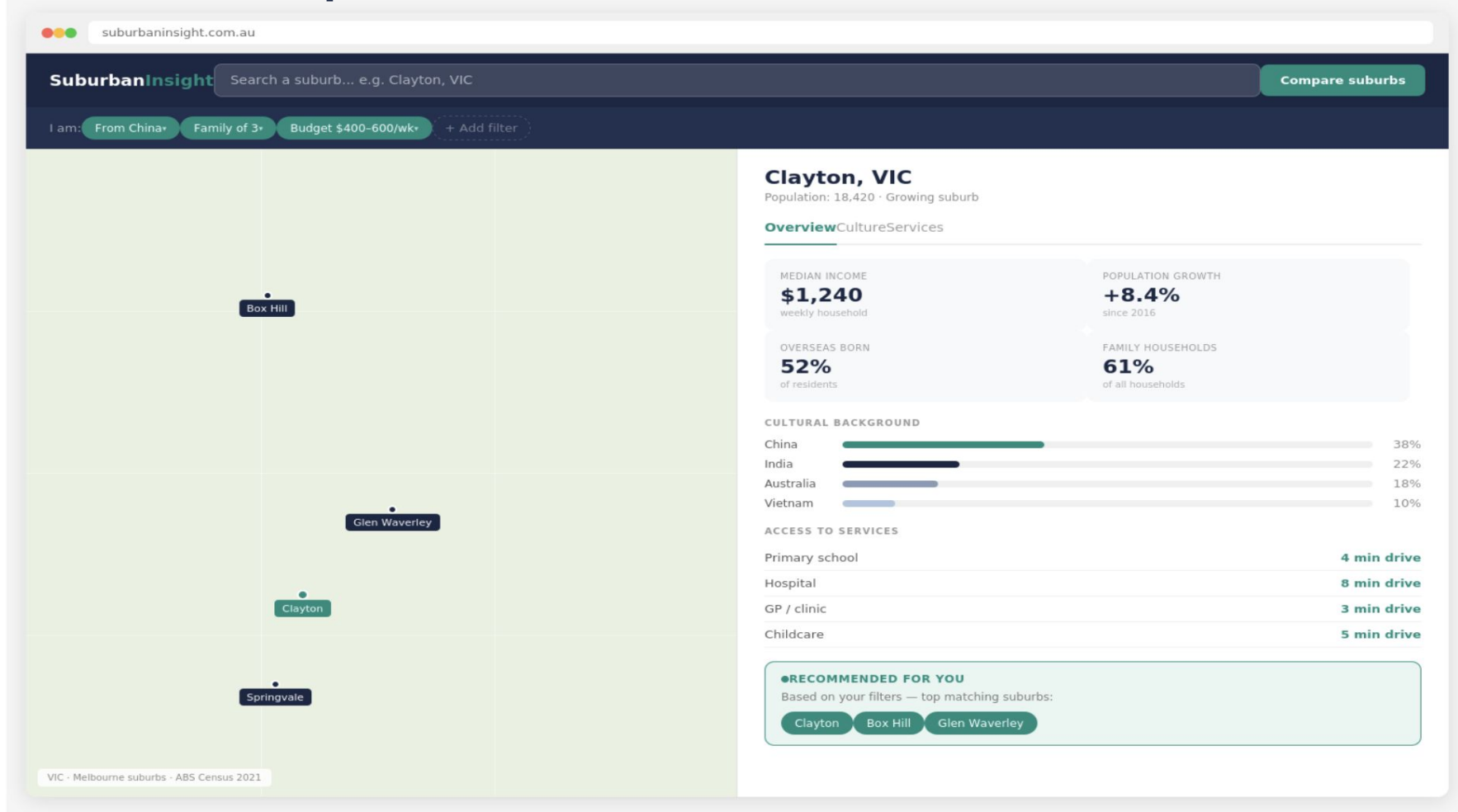
NFR, Med

Data accuracy from official ABS Census sources

Users making settlement decisions need trustworthy, credible data

Wireframe: Map & Search

Liuxiaoxi Yang

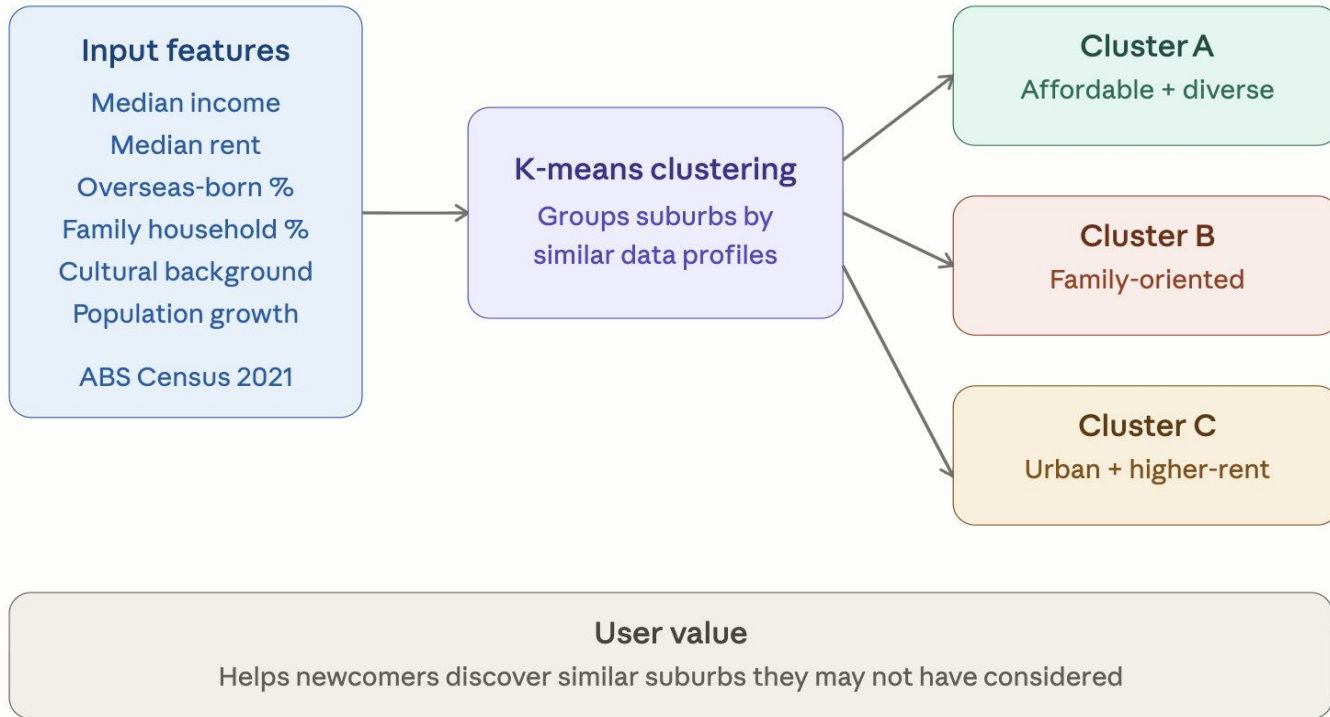


Wireframe: Machine Learning Breakdown

Liuxiaoxi Yang

How ML supports "recommended for you"

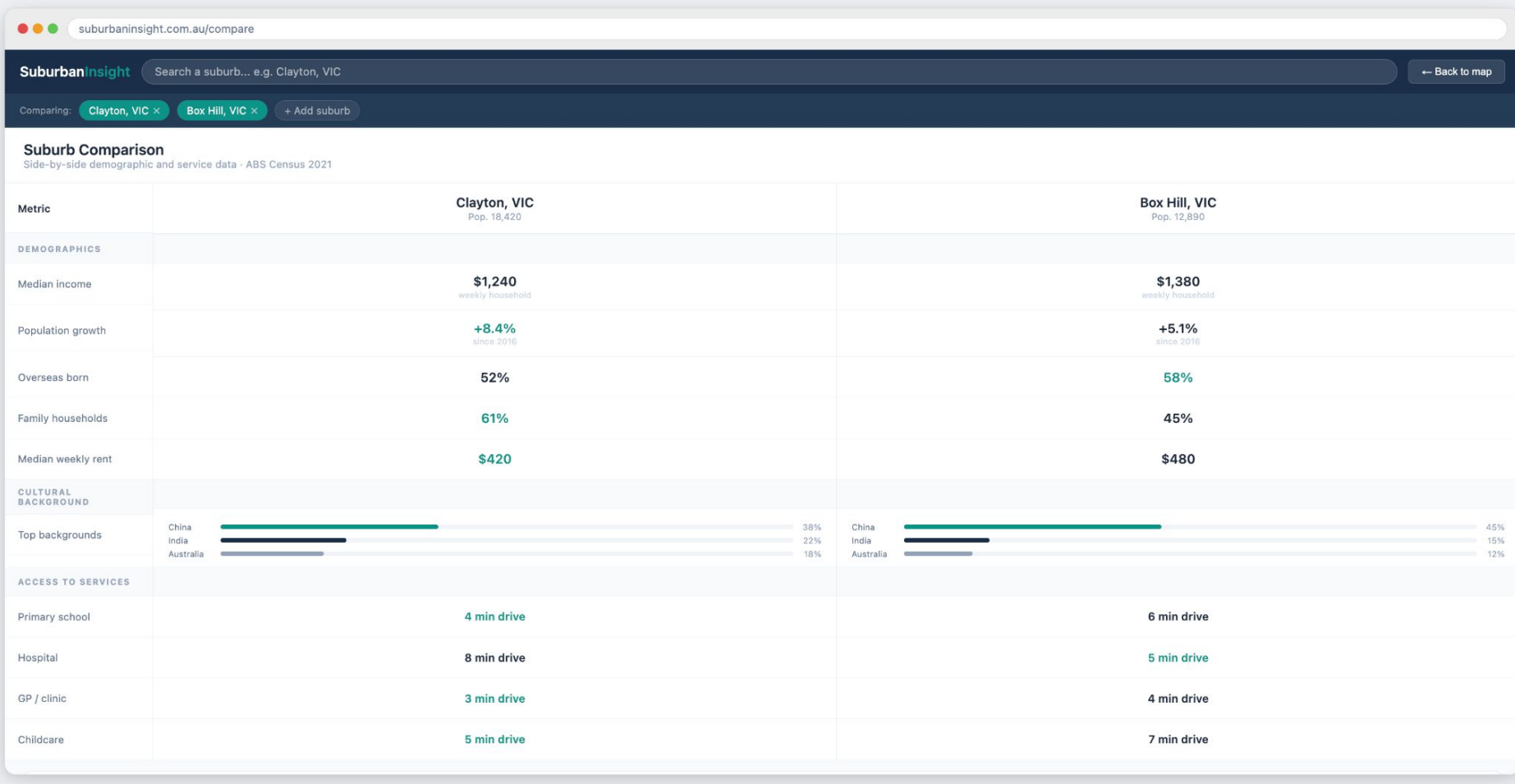
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Illustrative example only; final clusters will be generated from ABS data

Wireframe: Comparison in Suburbs

Liuxiaoxi Yang



Technical Stack & Feasibility

Jiajing Xu

FRONTEND

JavaScript · Leaflet.js · D3.js

Interactive map rendering, data visualisation, suburb comparisons

BACKEND

Python · FastAPI

ABS data processing, API endpoints, ML model serving

MACHINE LEARNING

scikit-learn · K-means · StandardScaler

Suburb clustering by demographic similarity

DATA

ABS Census 2021 · ABS Access Measures

Demographics, cultural data, drive time to services

DEPLOYMENT

Vercel (frontend) · Render (backend)

Free-tier hosting, no login, no installation required

Data, Assumptions & Constraints

Jiajing Xu

ASSUMPTIONS

- ABS Census data is publicly available for academic use
- Target users have basic computer literacy and desktop browser access
- Prototype scoped to Melbourne suburbs
- Semester 1 = research and design only; implementation in Semester 2

CONSTRAINTS

- No budget for paid APIs or cloud services
- Team of 4 students with varied web and data skills
- Must use publicly available data sources only (ABS)
- Out of scope: real-time data, mobile app, user accounts

Testing Approach

Jiajing Xu

- Functional testing: verify all API endpoints return correct data; all UI interactions work end-to-end
- Cross-browser testing: ensure compatibility across Chrome, Firefox, and Safari
- Performance testing: map load time and API response time within acceptable range
- Usability testing: recruit 5+ international students to complete key tasks (find suburb, apply filters, compare suburbs)
- Evaluation metrics: task completion rate, navigation clarity, data comprehension accuracy, user satisfaction



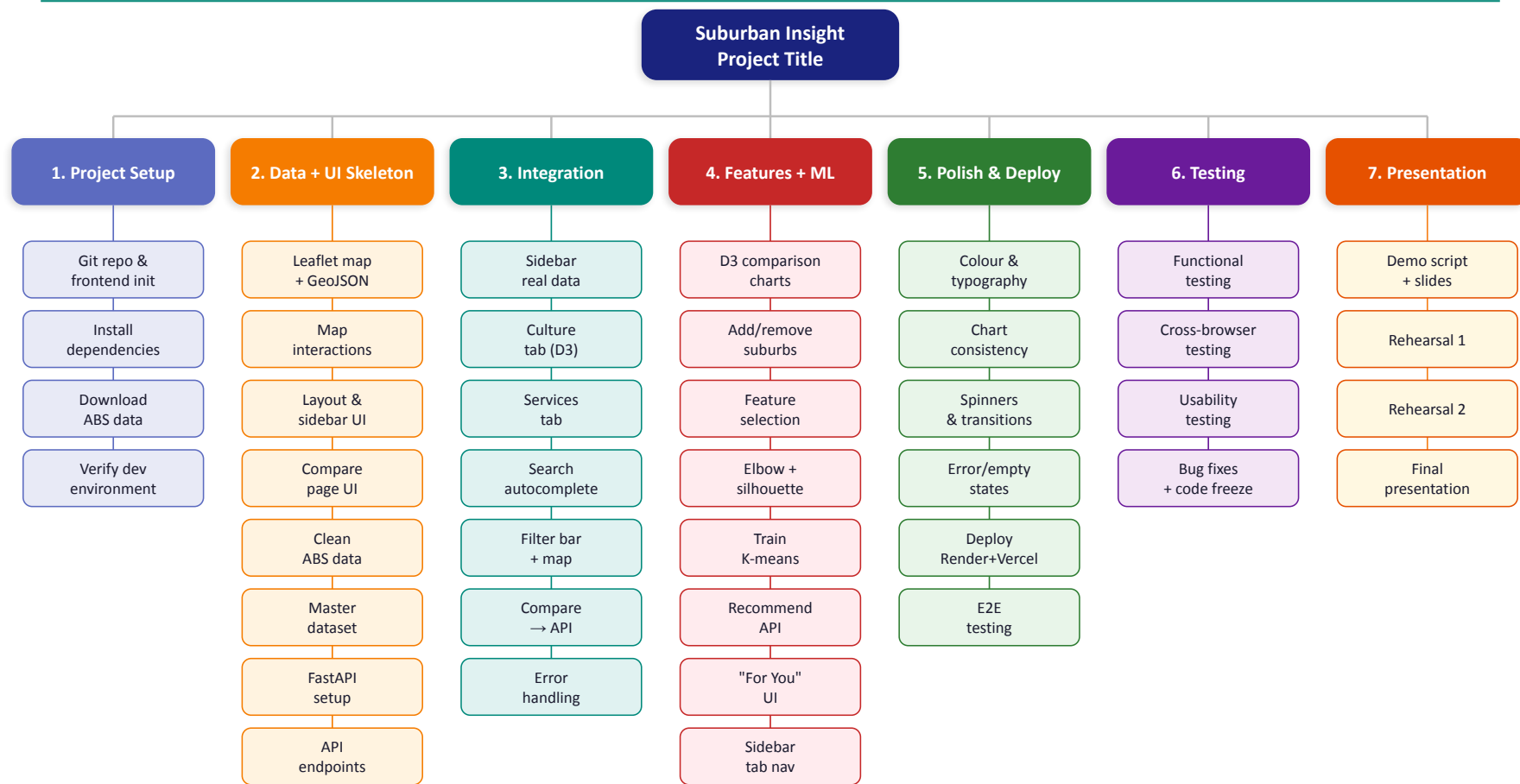
Project Management Methodology

Aidan Kwok

- Agile approach with iterative sprint cycles aligned to fortnightly unit milestones
- Regular team meetings via Zoom for progress check-ins, task allocation, and decision-making
- Shared documents for collaboration on deliverables (RTM, meeting minutes, presentation slides)
- Sprint reviews at each milestone: Week 5 pitch, Week 9 RTM draft, Week 10 final RTM, Week 11 presentation
- Semester 2: two-week sprints with milestones at W4 (MVP), W7 (Feature Complete), W10 (Final Build)

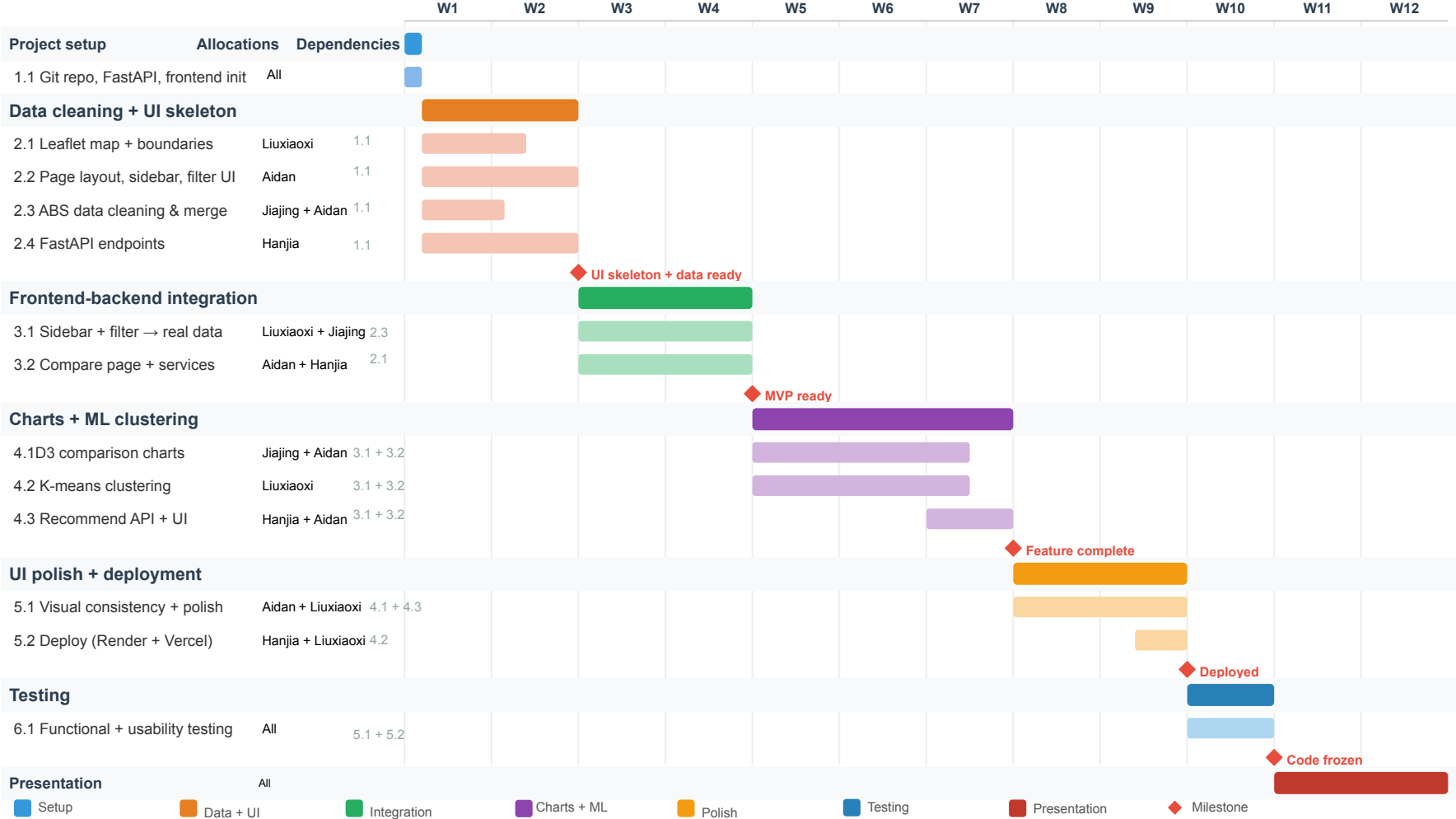
Work Breakdown Structure (WBS)

Aidan Kwok



Gantt Chart

Hanjia Liu



Conclusion

Hanjia Liu

IN SCOPE

- Interactive suburb map with ABS demographics, filtering, comparison, ML clustering, web deployment

OUT OF SCOPE

- No Real-time data, no mobile app, no non-ABS sources, no user login system, no paid APIs

Anticipated Challenges

- ABS data cleaning complexity
- ML clustering may not produce meaningful groups



Effective Teamwork

- Clear role allocation and regular communication helped our team work more efficiently

Effective Project Management

- Breaking the project into smaller phases and setting clear milestones helped the team stay organised and manage the workload more effectively

References

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AI Acknowledge

I (Liuxiaoxi) used Claude (www.claude.ai) in this presentation

UI Mockups — Generated two wireframe mockups (map view and comparison view) for slides 7–8.

Prompt: "Create a UI mockup for a web-based suburb explorer called Suburban Insight, targeting newcomers to Australia. The interface should include a search bar for suburbs, filter options for country of origin, family size, and weekly budget, an interactive map showing suburb locations with labels such as Clayton, Box Hill, Glen Waverley, and Springvale, and a sidebar displaying suburb details such as median income, population growth, overseas-born percentage, cultural background breakdown, and drive time to nearby schools, hospitals, GP clinics, and childcare. Include a 'Recommended for You' section using ML clustering results. Use a dark navy and teal colour scheme. Data is sourced from ABS Census 2021."

Thank You For Your Attention

We welcome any questions

Appendix

Detailed Gantt Chart:

<https://docs.google.com/spreadsheets/d/1CWZrt4EqDBHoyiRM8kWr62k6BTot8xE6/edit?usp=sharing&ouid=114065858801444372344&rtpof=true&sd=true>